

Theoretical Computer Science Exam, June 27th, 2017

The exam consists of **4 exercises**. Available time: 1 hr 20 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

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Exercise 1 (10 pts)

Consider the language $L = \{ x \in \{a, b\}^* \mid |x| \text{ is odd and the first, middle and last character of } x \text{ are equal} \}$

1. Write a context-free grammar that generates L
2. Design a push down automaton that accepts L
3. Argue that L is not regular

Exercise 2 (8 pts)

A procedure $sumArr(a, b, dim, p)$ has 3 input parameters a, dim, p , and one output parameter b . Parameters a and b are arrays of integers having dim elements, and it can be assumed that $dim \geq 3$, while parameter p can be assumed to be equal to either 0 or 1.

If $p=0$ then after the execution of $sumArr$ the first two elements of b are equal to the elements of a in the same positions, and every element from the third position on

- is equal to the sum of the two preceding elements if the second preceding element is greater or equal to the immediately preceding element;
- is equal to the product of the two preceding elements if the second preceding element is less than the immediately preceding element.

If $p=1$ then after the execution of $sumArr$ every element of b is equal to the sum all elements of a having position less than or equal to it.

For instance, if $a=[7 \ 3 \ 5 \ 6 \ 6 \ 9 \ 5]$, $dim=7$, $p=0$ then $b=[7 \ 3 \ 10 \ 15 \ 30 \ 12 \ 54]$, while

if $a=[7 \ 3 \ 5 \ 6 \ 6 \ 9 \ 5]$, $dim=7$, $p=1$ then $b=[7 \ 10 \ 15 \ 21 \ 27 \ 36 \ 41]$.

Specify the procedure $sumArr$ by means of a pre-condition and a post-conditions expressed in first-order logic with the usual relational and arithmetic operators typical of common programming languages (NB: no use of the summation operator like “ $\sum \dots$ ” is allowed).

Exercise 3 (8 pts)

Let M be a generic Turing machine, considered as a language acceptor, so that $L(M)$ denotes the language accepted by M . Answer the following questions, providing concise justifications.

1. Is the set $S_1 = \{ M \mid |L(M)| = 13 \}$ decidable?
2. Is the set $S_2 = \{ M \mid |L(M)| \geq 13 \}$ decidable? Is it semidecidable?
3. Is the set $S_3 = \{ M \mid |L(M)| \leq 13 \}$ decidable? Is it semidecidable?
4. Is the set $S_4 = \{ \langle M, x \rangle \mid x \text{ is accepted by } M \text{ in } \leq 13 \text{ steps} \}$ decidable? Is it semidecidable?

Exercise 4 (8 pts)

Consider the language $L = \{ a^n b^m c^{n \times m} \mid n, m > 0 \}$

1. design, preferably in full detail, or at least sketch in a precise and unambiguous way, a Turing machine that accepts L , and evaluate its time and space complexity;
2. design, preferably in full detail, or at least sketch in a precise and unambiguous way, a RAM machine that accepts L , and evaluate its time and space complexity both with reference to the uniform and the logarithmic cost criterion;
3. determine which of the two above computing models (TM and RAM) is more efficient, considering also the logarithmic cost criterion. Provide justifications for your answers.

Solutions

Exercise 1

Consider the language $L = \{ x \in \{a, b\}^* \mid |x| \text{ is odd and the first, middle and last character of } x \text{ are equal} \}$

1. Write a context-free grammar that generates L

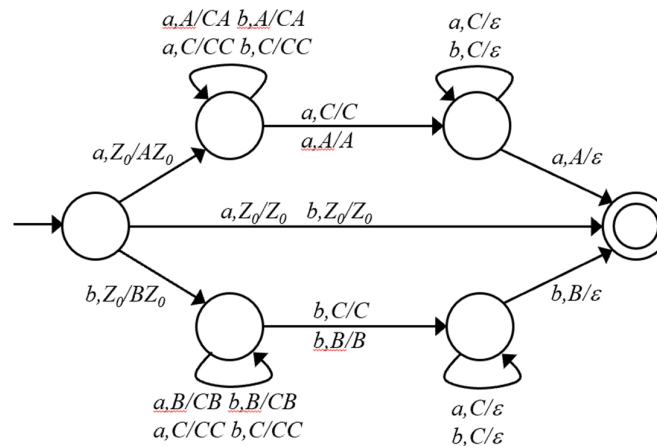
$S \rightarrow a \mid a A a \mid b \mid b B b$

$A \rightarrow a \mid a A a \mid a A b \mid b A a \mid b A b$

$B \rightarrow b \mid a B a \mid a B b \mid b B a \mid b B b$

2. Design a push down automaton that accepts L

A nondeterministic PDA is necessary, to guess the center of the string



3. Argue that L is not regular.

The automaton accepting L must identify (possibly nondeterministically) the central character, which requires an ability to count the length of strings that a finite state automaton does not possess. A more formal proof that L is not regular can be provided based on the Pumping Lemma.

Exercise 2

Pre: $\dim \geq 3 \wedge (p=0 \vee p=1)$

Post:

If $p=0$ then after the execution of *sumArr* the first two elements of b are equal to the elements of a in the same positions, and every element from the third position

- is equal to the sum of the two preceding elements if the second preceding element is greater or equal to the immediately preceding element;
- is equal to the product of the two preceding elements if the second preceding element is less than the immediately preceding element.

$(p=0 \rightarrow b[0]=a[0] \wedge b[1]=a[1] \wedge$

$\forall i (1 < i < \dim \rightarrow ((a[i-2] \geq a[i-1] \rightarrow b[i] = a[i-2] + a[i-1]) \wedge$

$(a[i-2] < a[i-1] \rightarrow b[i] = a[i-2] * a[i-1]) \vee \dots) \wedge$

If $p=1$ then after the execution of *sumArr* every element of b is equal to the sum all elements of a having position less than or equal to it.

$(p=1 \rightarrow b[0]=a[0] \wedge$

$\forall i (1 \leq i < \dim \rightarrow b[i] = b[i-1] + a[i]) \vee \dots)$

Exercise 3

Let M be a generic Turing machine, considered as a language acceptor, so that $L(M)$ denotes the language accepted by M . Answer the following questions, providing concise justifications.

1. Is the set $S_1 = \{ M \mid |L(M)| = 13 \}$ decidable?

No, as a consequence of the Rice theorem

2. Is the set $S_2 = \{ M \mid |L(M)| \geq 13 \}$ decidable? Is it semidecidable?

No, as a consequence of the Rice theorem; yes, by using the customary dovetailing technique.

3. Is the set $S_3 = \{ M \mid |L(M)| \leq 13 \}$ decidable? Is it semidecidable?

No, as a consequence of the Rice theorem; no, because the set $\{ M \mid |L(M)| > 13 \}$, of which S_3 is the complements, is undecidable and semidecidable (for the same reasons as S_2), therefore if S_3 was semidecidable both S_2 and S_3 would be decidable.

4. Is the set $S_4 = \{ \langle M, x \rangle \mid x \text{ is accepted by } M \text{ in } \leq 13 \text{ steps} \}$ decidable? Is it semidecidable?

S_4 is decidable: since the length of the computation is bounded it suffices to simulate the execution of M on x up to that number of steps.

Exercise 4

Consider the language $L = \{ a^n b^m c^{n \times m} \mid n, m > 0 \}$

1. design, preferably in full detail, or at least sketch in a precise and unambiguous way, a Turing machine that accepts L , and evaluate its time and space complexity;
2. sketch, in a precise and unambiguous way, a RAM machine that accepts L , and evaluate its time and space complexity both with reference to the uniform and the logarithmic cost criterion;
3. determine which of the two above computing models (TM and RAM) is more efficient, considering also the logarithmic cost criterion. Provide justifications for your answers.

The Turing machine reads the a symbols and copies them on a (first) memory tape, then it reads the b symbols and copies them on a (second) memory tape; the two memory tapes are then used to count the c symbols while reading them (e.g., the head on the first memory tape is moved once for every c symbol read from the input, and the head of the second memory tape is moved by one position only when the leftmost or rightmost position is reached on first memory tape).

Clearly a single pass on the input is sufficient, and the input is checked, in the worst case, in real-time (plus a small, constant number of moves). Therefore: $T(n) = \Theta(n)$. The worst case for space complexity is when $n=1$ or $m=1$; in that case, if x is the input then $S(|x|) = \Theta(|x|)$.

The RAM machine reads and counts the a 's and the b 's, then it computes the value $n \times m$, it reads and counts the c 's and finally checks that the number of symbols is correct. Therefore, with the uniform cost criterion, $S(n) = \Theta(1)$, $T(n) = \Theta(n)$; the RAM is superior to the Turing machine for what concerns space complexity, and it is comparable regarding time complexity. Adopting the logarithmic cost criterion, $S(n) = \Theta(\log n)$, $T(n) = \Theta(n \log n)$; the RAM is still superior to the Turing machine for what concerns space complexity, whereas time complexity is better for the Turing machine.